

MANAGEMENT OF CHOROIDAL MELANOMAS LESS THAN 10 mm IN LARGEST BASAL DIAMETER WITH A 10-mm RUTHENIUM PLAQUE

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Purpose: To assess tumor control, complications, and vision after brachytherapy for posterior choroidal melanoma <10 mm in largest basal diameter with the 10-mm ruthenium plaque.

Methods: This was a retrospective cohort study of consecutive choroidal melanomas <10 mm by largest basal diameter in a national ocular oncology service in 1998 to 2010. The median dose was 116 Gy (range, 80–194 Gy) to the apex and 327 Gy (range, 201–824 Gy) to the sclera. The median tumor height and largest basal diameter were 1.9 mm (range, 0.4–5.2 mm) and 7.0 mm (range, 3.3–9.6 mm), respectively. The median distance to disk and foveola was 3.0 mm (range, 0–7.5 mm) and 2.0 mm (range, 0–8.5 mm), respectively.

Results: Four recurrences occurred at a median of 1.4 years (range, 0.6–3.1 years) after irradiation. Five-year cumulative incidence of local recurrence was 9% (95% confidence interval [CI], 3–20). Six patients died at a median of 4.2 years (range, 0.28–8.6 years) after treatment, one with evidence of metastases. At 5 years, 57% (95% CI, 31–79), 72% (95% CI, 58–85), and 97% (95% CI, 88–100) of eyes were free of any maculopathy, radiation maculopathy, and optic neuropathy, respectively. Cumulative incidence of developing low vision and blindness were 17% (95% CI, 7–31) and 3% (95% CI, 2–12) at 5 years, respectively. Thickness >3.0 mm, largest basal diameter >7.0 mm, and location ≤1.5 mm of foveola were associated with visual loss.

Conclusion: Local tumor control and vision outcomes support the use of 10-mm ruthenium plaques in managing smallest choroidal melanomas.

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Early recognition of small choroidal melanomas is important because 12% of them lead to clinical metastases within 10 years.¹ Characteristics predictive of growth and malignancy of small choroidal melanocytic tumors include thickness >2 mm, subretinal fluid, symptoms, orange pigment, and tumor margin touching or located within 3 mm of the optic disk.^{2,3} These five features help decide which tumors are of high risk and should be managed as melanomas.

Enucleation is no longer a widely accepted alternative when managing small tumors. Small malignant choroidal melanomas are typically managed conservatively, often allowing the preservation of useful vision. The primary aims are to destroy the tumor and to prevent local recurrences, which might be associated with an increased risk for metastases.⁴ Several

options are available for conservative managing of small choroidal melanomas. Most are treated either with episcleral plaque brachytherapy alone or combined with transpupillary thermotherapy (TTT).⁵ Choroidal melanomas are relatively resistant to irradiation, and the doses used often lead to radiation-related complications.

Various types of isotope and plaque are available. A 10-mm ruthenium-106 plaque (Ru-CCX) has been the treatment of choice for smallest choroidal melanomas at the Department of Ophthalmology, Helsinki University Central Hospital, since 1998. Compared with a 15-mm plaque, which is more widely used, the irradiated volume is smaller, and complication rates are potentially lower. We report local tumor control, complications, and vision in 45 consecutive patients

with a posterior choroidal melanoma <10 mm in diameter managed with the Ru-CCX plaque.

Methods

Aim

Our aim was to assess local tumor control, complications, visual acuity, and mortality in a population-based consecutive series of patients with a choroidal melanoma <10 mm by largest basal tumor diameter (LBD) who underwent brachytherapy with a 10-mm ruthenium plaque.

Inclusion Criteria

Eligible were patients diagnosed with a choroidal melanoma <10 mm in LBD between October 1998 and December 2010 and scheduled to have a 10-mm Ru-CCX plaque. The LBD was based on indirect ophthalmoscopy and B-scan echography. We reviewed the database of the Ocular Oncology Service, Department of Ophthalmology, Helsinki University Central Hospital (HUCH), a national referral center for uveal melanoma patients, and identified 118 with uveal melanomas <10 mm in LBD, of whom 48 (41%) had been scheduled for Ru-CCX plaque brachytherapy. Three had a tumor that involved the ciliary body or iris. This group is compared with the 70 tumors scheduled for other treatments provided as **Supplemental Digital Contents 1 and 2** (see **Figure** and **Table**, <http://links.lww.com/IAE/A242> and <http://links.lww.com/IAE/A243>, which show types of various plaques used and tumor dimensions by plaque type).

Forty-four of the patients had been scheduled for brachytherapy as primary treatment and one as a secondary treatment for regrowth 20 years after 3 sessions of argon laser photocoagulation. One patient eventually received a ruthenium-CCA plaque because it was impossible to suture the Ru-CCX plaque in

place because of posterior tumor location and a tight orbit. We included this patient in all analyses in accordance with the intention-to-treat principle. The study followed the tenets of the Declaration of Helsinki, and data analysis was retrospectively approved by the institutional review board.

Clinical Evaluation

We measured the best-corrected visual acuity using a test-type projector (Rodavist 2 and 524; Rodenstock GmbH, Ottobrunn, Germany) and even decimals from 1.0 (20/20) to 0.1 (20/200) plus 0.05 (20/400). Visual acuity worse than 20/400 was recorded as counting fingers at 2, 1, and 0.5 m, hand movement, light perception, and no light perception. We used biomicroscopy and indirect ophthalmoscopy to localize the tumor margins, to estimate the distance from the posterior tumor margin to the optic disk and foveola, and to evaluate its growth pattern, presence of subretinal fluid, and presence of orange pigment.

We treated the patients if they had a presumed small choroidal melanoma, which either had been observed to grow (18 [40%] patients) or had TFSOM (To Find Small Ocular Melanomas)^{2,3} high-risk characteristics predictive of growth (Table 1). The range of observed growth was 0.1 mm to 2.3 mm in thickness and 0.1 mm to 5.4 mm in LBD. The high-risk characteristics included thickness >2 mm (18 [40%]), subretinal fluid (28 [62%]), symptoms (25 [56%]), orange pigment (28 [62%]), and margin within 2 disk diameters [DD] of the optic disk (26 [58%], of which 3 [7%] touched the disk margin).^{2,3} Symptoms included decreased vision, a visual field defect, flashes, or floaters. Other risk characteristics recorded retrospectively were ultrasound hollowness (32 [71%]), absence of

Table 1. Number of Risk Characteristics for Growth in 45 Patients With Small Choroidal Melanoma

Number of Risk Characteristics	Posterior Tumor Margin		
	Within 2 DD of Disk*, n (%)	Touching Disk†, n (%)	Observed Growth‡, n (%)
0	4 (9)	4 (9)	4 (100)
1	5 (11)	10 (22)	8 (80)
2	9 (20)	9 (20)	4 (44)
3	11 (24)	14 (31)	4 (29)
4	11 (24)	8 (18)	1 (12)
5	5 (11)	0 (0)	

Melanoma 10 mm or less in LBD managed with the Ru-CCX plaque.

*Later TFSOM high-risk criteria.³

†Original TFSOM high-risk criteria.²

‡Before or after brachytherapy, percentage is calculated for tumors fulfilling the original TFSOM criteria.

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surrounding halo (44 [98%]), and absence of drusen (18 [40%]).⁵ Four of the patients treated because of observed growth had none of the risk characteristics.

Brachytherapy

The Ru-CCX plaque (Eckert & Ziegler BEBIG GmbH, Berlin, Germany) has a physical diameter of 11.6 mm and a radius of curvature of 12 mm. The diameter of the radiation window is 10 mm before reshaping the applicator to spherical form. It is surrounded by a 0.75-mm-wide inactive border. In the spherical form, the cord diameter of the radiation window is 9.6 mm. This is surrounded by a penumbra of scatter radiation, which was advantageous in eccentric treatments^{6,7} and in treating selected tumors up to 9.6 mm in diameter.

The prescription point was tumor thickness plus 1 mm for the sclera. The standard prescription dose was either 100 Gy (20 patients) or 120 Gy (19 patients) to the apex depending on the thickness of the tumor, aiming to deliver a minimum dose of 250 Gy to the sclera. The median delivered dose (Figure 1A) was 116 Gy (range, 80–194 Gy) to the apex and 327 Gy (range, 201–824 Gy) to the sclera over a median of 93 hours (range, 28–266 hours). We detached the inferior oblique muscle in 17 eyes in which the tumor was located in the macular area. Details of brachytherapy are provided as **Supplemental Digital Content 3** (see **Text**, <http://links.lww.com/IAE/A244>, including deviations from the standard doses and surgical technique).

Assessment of Local Tumor Control, Vision, and Complications

We followed patients prospectively at 3, 6, and 12 months and every 6 months until 3 years to 4 years from irradiation. Thereafter, 14 of them were seen annually in a central hospital elsewhere, 9 in other universities, and the rest in our service. We used biomicroscopy, indirect ophthalmoscopy, fundus photography, and B-scan echography (I³ System-ABD; Innovative Imaging, Sacramento, CA) to assess tumor control. We coded recurrence as vertical, marginal, or extrascleral.¹⁰ An unequivocal change of 0.5 mm or more in diameter or 0.3 mm in thickness indicated a marginal and vertical recurrence, respectively.

We retrospectively modeled the location of the plaque to facilitate determining the reasons for local recurrence. We used the surgical notes together with the scars from the diathermy markings and radiation, if visible, to draw digitally a 9.6-mm circle corresponding to the radiation window on a color fundus photograph, presuming the average diameter of the optic disk to be 1.5 mm. We coded the location of the

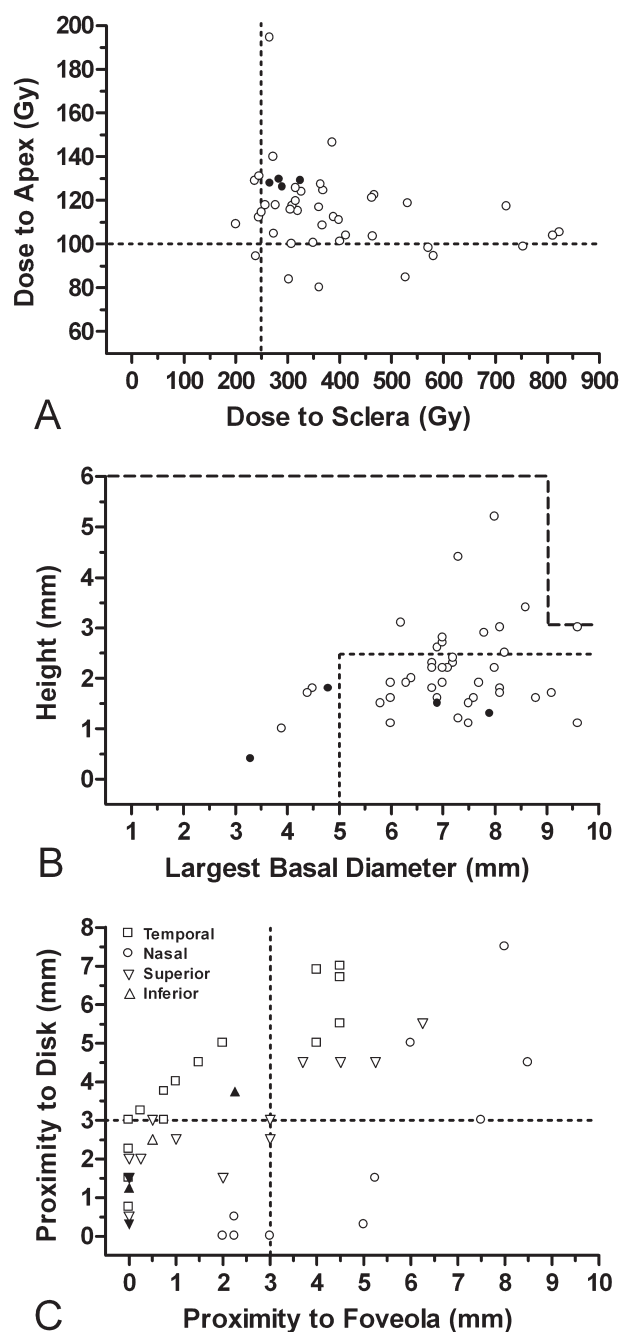


Fig. 1. Characteristics of 45 small choroidal melanomas <10 mm in largest basal tumor diameter (LBD) irradiated with the 10-mm ruthenium CCX plaque. **A.** Scatterplot of radiation dose to the sclera against dose to the tumor apex. Dashes indicate minimum prescription doses. **B.** Scatterplot of LBD against tumor height. Long and short dashed lines enclose tumors, which fulfill the American Joint Committee on Cancer (AJCC) Tumor, Node, Metastasis (TNM, seventh edition),⁸ and the Collaborative Ocular Melanoma Study (COMS)⁹ criteria for a small uveal melanoma, respectively. **C.** Scatterplot of proximity of tumor margin to the foveola against proximity to the optic disk. Dashes indicate a distance of 2 DD (3 mm) from these structures, often considered to be a safety margin for not developing radiation maculopathy and optic neuropathy from ruthenium brachytherapy. **A–C.** Solid and open symbols indicate tumors with and without recurrence, respectively.

plaque relative to tumor margins as covering the entire tumor with safety margin, being parallel with at least one tumor margin, or not covering the entire tumor.

We assessed the visual outcome after conventions of the Collaborative Ocular Melanoma Study (COMS)¹¹ and complications and cosmesis as outlined in a previous study from our service.¹²

Assessment of Metastatic Status

We performed liver function tests and upper abdominal ultrasonography at the time of diagnosis and thereafter annually, and a chest radiogram at diagnosis. Details of classifying survival outcomes are provided as **Supplemental Digital Content 3** (see **Text**, <http://links.lww.com/IAE/A244>, including conventions of classifying the cause of death).

Statistical Methods

We collected follow-up data up to date of analysis on November 30, 2011, prospectively into a dedicated database (MS Access; Microsoft Co, Seattle, WA) and analyzed them with Stata statistical software (Release 10; Stata Co, College Station, TX). We plotted the percentage regression in tumor thickness at 6-month intervals. We choose cumulative incidence instead of Kaplan–Meier analysis to estimate the time to individual complications as primary analysis. The Kaplan–Meier method assumes that all patients censored from analysis are still at risk of the event of interest.¹³ When used to analyze events other than death, the Kaplan–Meier method provides biased estimates if any patients are censored from the analysis because of a competing risk event, which renders them immune to the event of interest.¹³ We modeled death as a competing risk event, after the occurrence of which the patient was no longer at risk of recurrence or complications. Cumulative incidences can be summated and subtracted from 1 to obtain an estimate of the proportion of patients who survive without complications. Patients were censored from the analysis at the end of the follow-up.

We secondarily assessed time to local recurrence, time to development of complications, time to visual loss, and all-cause mortality by Kaplan–Meier analysis to compare estimates of the 5-year incidence of complications (see **Table**, **Supplemental Digital Content 4**, <http://links.lww.com/IAE/A245>, comparing cumulative incidence and Kaplan–Meier estimates).

The mean visual acuity, after transformation to a logarithmic scale and stratified by factors associated with poor vision outcomes, was plotted at 6-month intervals. Measurements below 20/1,600, such as counting fingers or hand movement, were set equal

to 20/1,600. Enucleated eyes and eyes with no light perception were counted as having 20/2,500 vision.

We analyzed the development of low vision (visual acuity 20/70 or worse) and blindness (visual acuity <20/400) using the World Health Organization (WHO) standard definition of low vision and blindness¹⁴ and the cumulative incidence method. Loss of vision was recorded when confirmed at the next review examination. If a measurement had not been made at every review, we considered the visual acuity to be the average of the previous and the next visit. Otherwise, we coded the data as missing.

Results

The median age at diagnosis of the 45 enrolled patients (male:female, 20:25), all whites, was 61 years (range, 26–88 years; Table 2). The median follow-up time both for all patients and for those who were still alive was 5.2 years (range, 3.4 months–11.9 years).

Baseline Tumor Characteristics

All tumors were classified as T1aN0M0 (small), Stage 1, by the American Joint Committee on Cancer (AJCC) Tumor, Node, Metastasis (TNM) System, seventh edition⁸ (Figure 1B). The criteria of T1a are thickness ≤6.0 mm if LBD is ≤9.0 mm and thickness ≤3.0 mm when LBD is 9.1 mm to 12.0 mm. By COMS criteria, 5 tumors <5.0 mm in LBD would have been ineligible even to the small tumor observational study, 29 tumors were small, and 11 were medium sized.⁹

The median tumor thickness and LBD were 1.9 mm (mean, 2.1 mm; range, 0.4–5.2 mm) and 7.0 mm (mean, 7.0 mm; range, 3.3–9.6 mm), respectively. The posterior tumor margin touched the optic disk (juxtapapillary) in 3 eyes (7%) and extended to within 2 DD of the disk in 23 eyes (51%) (Table 2; Figure 1C). The median distance to the optic disk was 3 mm (mean, 3.1 mm; range, 0–7.5 mm). The posterior margin involved the foveola in 13 (29%) eyes and extended to within 2 DD of the foveola (perifoveal) in 30 (67%) eyes. The median distance to the foveola was 2.0 mm (mean, 2.5 mm; range, 0–8.5 mm). Ten tumors were nasal and 17 were temporal (Figure 1C). All but three were dome or flat in shape.

At the time of diagnosis, the median visual acuity was 20/25. Fourteen (31%) eyes had 20/20 vision or better, and 1 eye had worse than 20/200 vision (Table 2). By WHO criteria,¹⁴ 6 (13%) eyes had low vision and 1 was blind.

Cataract was present at the time of diagnosis in 12 (27%) eyes, and 5 (11%) had an intraocular lens. The

Table 2. Baseline Characteristics of 45 Patients With Small Choroidal Melanoma 10 mm or Less in largest basal tumor diameter (LBD) Managed With the Ru-CCX Plaque

Characteristic	n (%)
Age, years	
<40–49	10 (22)
50–59	12 (27)
60–69	11 (24)
70–>80	12 (27)
Gender	
Male	19 (42)
Female	26 (58)
Apical thickness, mm	
≤1.0–2.0	26 (58)
2.1–3.0	15 (33)
3.1–6.0	4 (9)
LBD, mm	
≤4.0–6.0	9 (20)
6.1–8.0	27 (60)
8.1–10.0	9 (20)
Anterior border	
Posterior to equator	44 (98)
Anterior to equator	1 (2)
Proximity to the optic disk, mm	
0.0–2.0	16 (36)
2.1–4.0	15 (33)
4.1–6.0	10 (22)
≥6.1	4 (9)
Proximity to the center of foveola, mm	
0.0–2.0	24 (53)
2.1–4.0	9 (20)
4.1–6.0	8 (18)
≥6.1	4 (9)
Shape	
Dome or flat	42 (93)
Peaked, irregular, or indeterminate	3 (6)
Subretinal fluid	
None noted	17 (38)
Over the tumor	9 (20)
Around the tumor	18 (40)
Exudative retinal detachment	1 (2)
Vitreous hemorrhage	3 (7)
Visual acuity, tumor eye	
≥20/20	14 (31)
20/25–20/32	10 (22)
20/40–20/80	8 (18)
≤20/200	2 (4)

intraocular pressure exceeded 21 mmHg in 2 (4%) of 42 eyes with known intraocular pressure at diagnosis, and 3 eyes (6%) had previous glaucoma not related to the tumor. Subretinal fluid was present over or around the tumor in 27 eyes (60%) at diagnosis (Table 2). An exudative retinal detachment extending to the ora serrata was present in one eye. Thirteen tumors directly involved the foveola, seven were associated with subretinal fluid involving the foveola, and one eye each had dry senile macular degeneration, an epiretinal membrane, and diabetic maculopathy. Vitreous hemorrhage was present in 3 eyes (7%).

Location of the Plaque

According to the surgical charts, the plaque covered the tumor with a safety margin in 13 (29%) eyes, was eccentric, that is, parallel with a tumor margin in 20 (44%) eyes or did not entirely cover a margin in 10 (22%) eyes, and was indeterminate in 2 (4%) eyes. By retrospective modeling (see **Figure, Supplemental Digital Content 5**, <http://links.lww.com/IAE/A246>, showing examples of retrospective modeling), the corresponding figures were 6 (13%), 13 (29%), 18 (40%), and 7 (16%) eyes. Modeling suggested that one plaque may have tilted along the optic nerve and thus may not have touched the sclera.

Tumor Regression and Local Recurrence

By 3 years of follow-up, the median tumor thickness had reduced by 48% (Figure 2A). A marginal local recurrence was detected in 4 patients (Figure 3), and 2 of these recurrences were at the same time vertical (i.e., diffuse recurrences¹⁰; details of all recurrences are provided as **Supplemental Digital Content 6** [see **Text**, <http://links.lww.com/IAE/A247>], including detailed case histories), all within 3 years 1 month from treatment. The cumulative incidence for developing local recurrence was 7% (95% CI, 2–17) and 9% (95% CI, 3–20) by 3 years and 5 years, respectively (Figure 2B). All were controlled with additional brachytherapy: 2 with a 10-mm iodine (I-CCX) and 2 with a 20-mm optic nerve notched iodine (I-COB) plaque.

Survival

Of the 45 patients, 6 died during the follow-up. Although not histopathologically verified, one of the deaths was classified as being the result of solitary metastatic choroidal melanoma to the brain after local tumor recurrence (for details, see **Text, Supplemental Digital Content 6**, <http://links.lww.com/IAE/A247>). Four patients died of cardiovascular disease, which in one case was certain and in others was probable, and 1 died of probable aspiration pneumonia at the age of 75 years to 90 years. The cumulative incidence estimate of all-cause mortality was 2% (95% CI, 3–15) by 3 years and 11% (95% CI, 4–26) by 5 years (Figure 2C).

Treatment-Related Complications

Cataract. The cumulative incidence of a cataract developing or progressing was 21% (95% CI, 10–30) by 3 years and 28% (95% CI, 14–43) by 5 years (Figure 2D). No lens opacity was classified as being the result of radiation.

Iris neovascularization and glaucoma. None of the eyes developed iris neovascularization. One patient

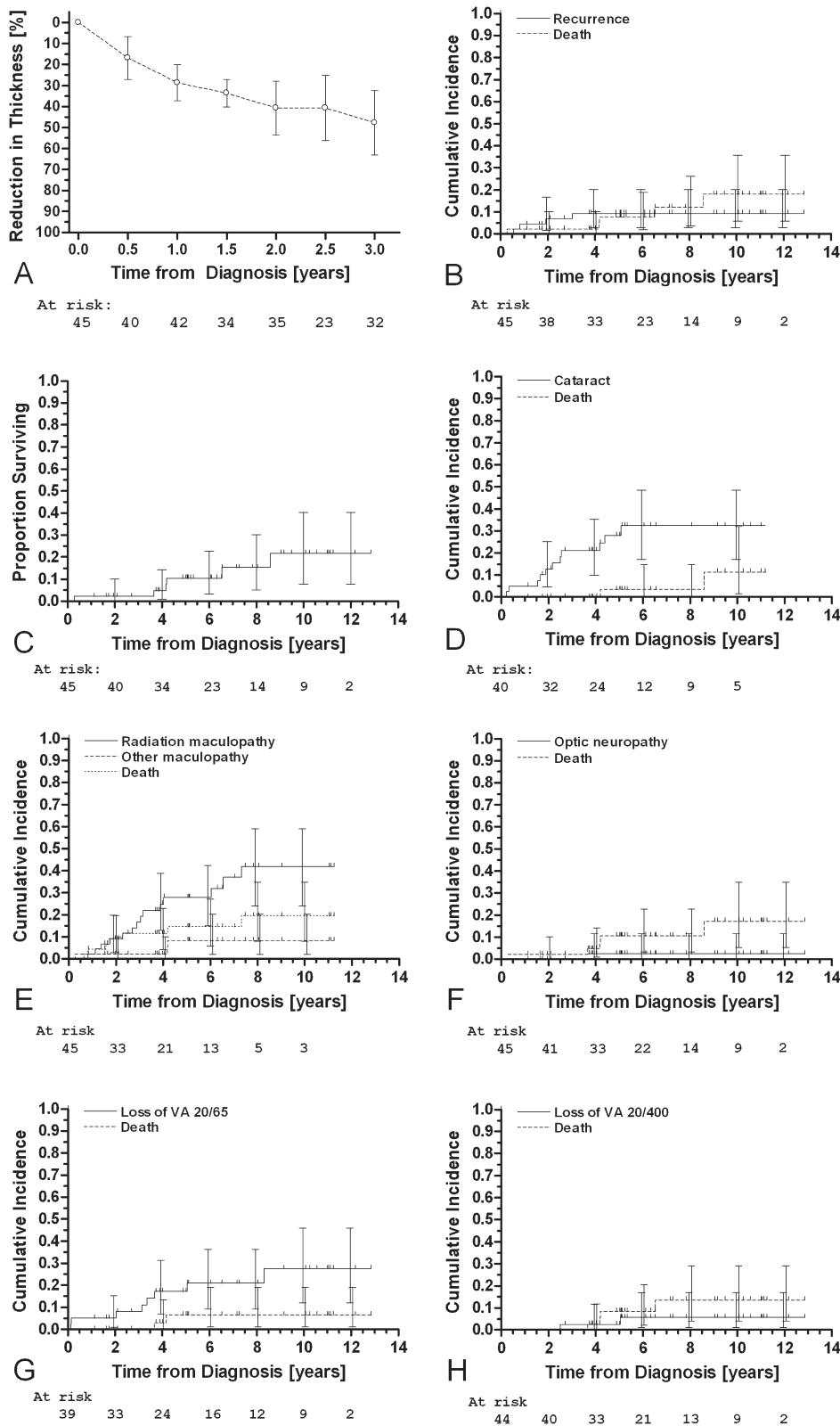
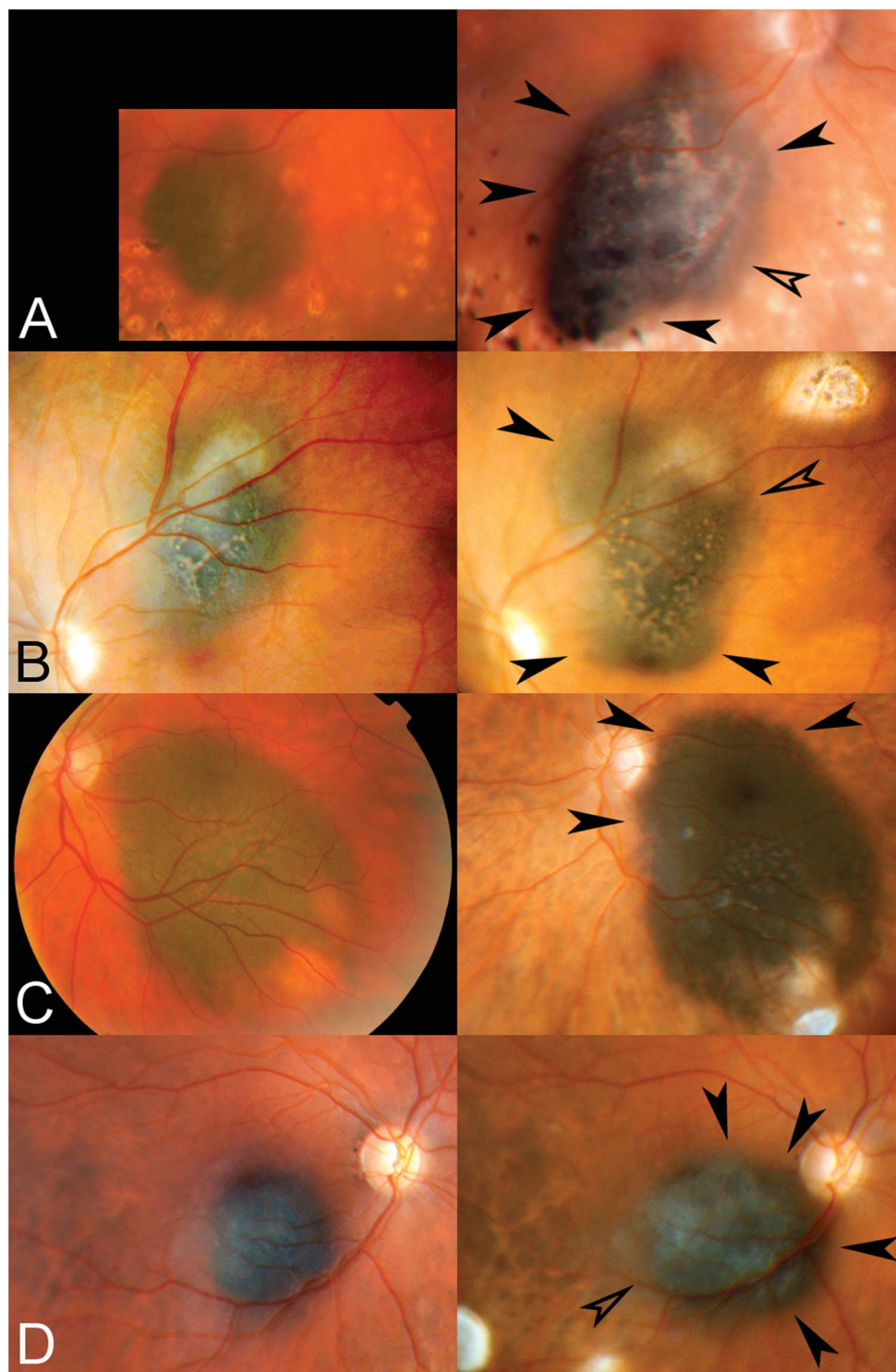


Fig. 2. Percentage reduction in tumor thickness after brachytherapy for 45 patients with a small choroidal melanoma irradiated with the ruthenium CCX plaque (**A**). Cumulative incidence of time to first local recurrence (**B**), death of any cause (**C**), cataract (**D**), maculopathy (**E**), optic neuropathy (**F**), and first occurrence of low vision (**G**), and blindness in the study eye (**H**) by World Health Organization (WHO) definitions. **B, D–H.** Cumulative incidence of death of any cause, the main competing risk, is also plotted. Crossbars show 95% CIs and ticks censored observations. Numbers below graphs are patients at risk. VA, visual acuity.

Fig. 3. The 4 recurrent tumors at the time of treatment (left) and local recurrence (right): Case 1 (A), Case 2 (B), Case 3 (C), and Case 4 (D). Open arrows indicate no observed growth, and solid arrows indicate growth (case reports of these tumors are provided as **Supplemental Digital Content 6** [see Text, <http://links.lww.com/IAE/A247>]).



was diagnosed with primary open-angle glaucoma 18 months after irradiation at the age of 77 years, and 1 patient experienced transient increase of intraocular pressure after irradiation.

Persistent exudative retinal detachment. None of the eyes developed a persistent exudative detachment.

Maculopathy. Maculopathy was diagnosed in 22 (49%) eyes after irradiation, of which 14 (31%) developed radiation maculopathy (see **Table, Supplemental Digital Content 7**, <http://links.lww.com/IAE/A248>, which includes the causes, treatment, and outcome of maculopathy). Three (7%) eyes developed a new

epiretinal membrane: a recurrent tumor extended to the foveola in 2 (4%) eyes, the radiation scar involved the fovea in one eye, and tumor-related lipid exudation developed in 1 eye. The cumulative incidence of developing radiation maculopathy was 14% (95% CI, 6–26) by 3 years and 28% (95% CI, 15–42) by 5 years (Figure 2E). The 3- and 5-year cumulative incidence of developing maculopathy other than radiation maculopathy were 12% (95% CI, 4–23) and 15% (95% CI, 6–27), respectively (Figure 2E).

Optic neuropathy. One eye developed radiation optic neuropathy 3 years 8 months after irradiation. The tumor was located 4 mm from the optic disk. The cumulative incidence of developing optic neuropathy was 3% (95% CI, 0–12) by 5 years (Figure 2F).

Vitreous hemorrhage. Two eyes developed vitreous hemorrhage after brachytherapy. The first one had a minor hemorrhage 4 months after irradiation, which resolved in a year. The second one first had a minor hemorrhage 3 years 7 months after irradiation, which progressed to moderate but gradually resolved during the next 4 years. The 3- and 5-year cumulative incidence of developing vitreous hemorrhage were 2% (95% CI, 0–15) and 5% (95% CI, 1–18), respectively.

Preservation of Visual Acuity

Three and 5 years after irradiation, the median visual acuity in the study eye was 20/30 and 20/25, respectively. The cumulative incidences of developing low vision and blindness by WHO criteria¹⁴ in the study eye 3 years after diagnosis were 8% (95% CI, 2–19) and 3% (95% CI, 2–12) of eyes that had better visual acuity at diagnosis, respectively. The corresponding percentages 5 years after diagnosis were 17% (95% CI, 7–31) and 3% (95% CI, 2–12), respectively (Figure 2, G and H).

Three years after diagnosis, the visual acuity was the same as the pretreatment visual acuity in 5 (15%; 95% CI, 5–32) of 33 eyes, better in 17 (51%; 95% CI, 34–69) eyes, and worse in 11 (33%; 95% CI, 18–52) eyes. At this time, it was within 1 line of pretreatment visual acuity in 23 eyes (70%; 95% CI, 51–84), at least 2 lines better in 3 eyes (9%; 95% CI, 2–24), and at least 2 lines worse in 7 eyes (21%; 95% CI, 9–39).

Characteristics Predicting Visual Loss

Age or preoperative visual acuity was not associated with a change in the mean visual acuity over time (Figure 4, A and B). Loss of vision was most notable when tumor thickness was >3.0 mm (Figure 4C) and when LBD was >7.0 mm (Figure 4D). Location within 1.5 mm of the optic disk (Figure 4E) was not associated with vision loss. Location within 1.5 mm of

the foveola was a strong indicator for losing visual acuity (Figure 4F).

Enucleation and Cosmetic Outcome

No secondary enucleation was performed. All eyes had normal cosmesis at the latest follow-up.

Discussion

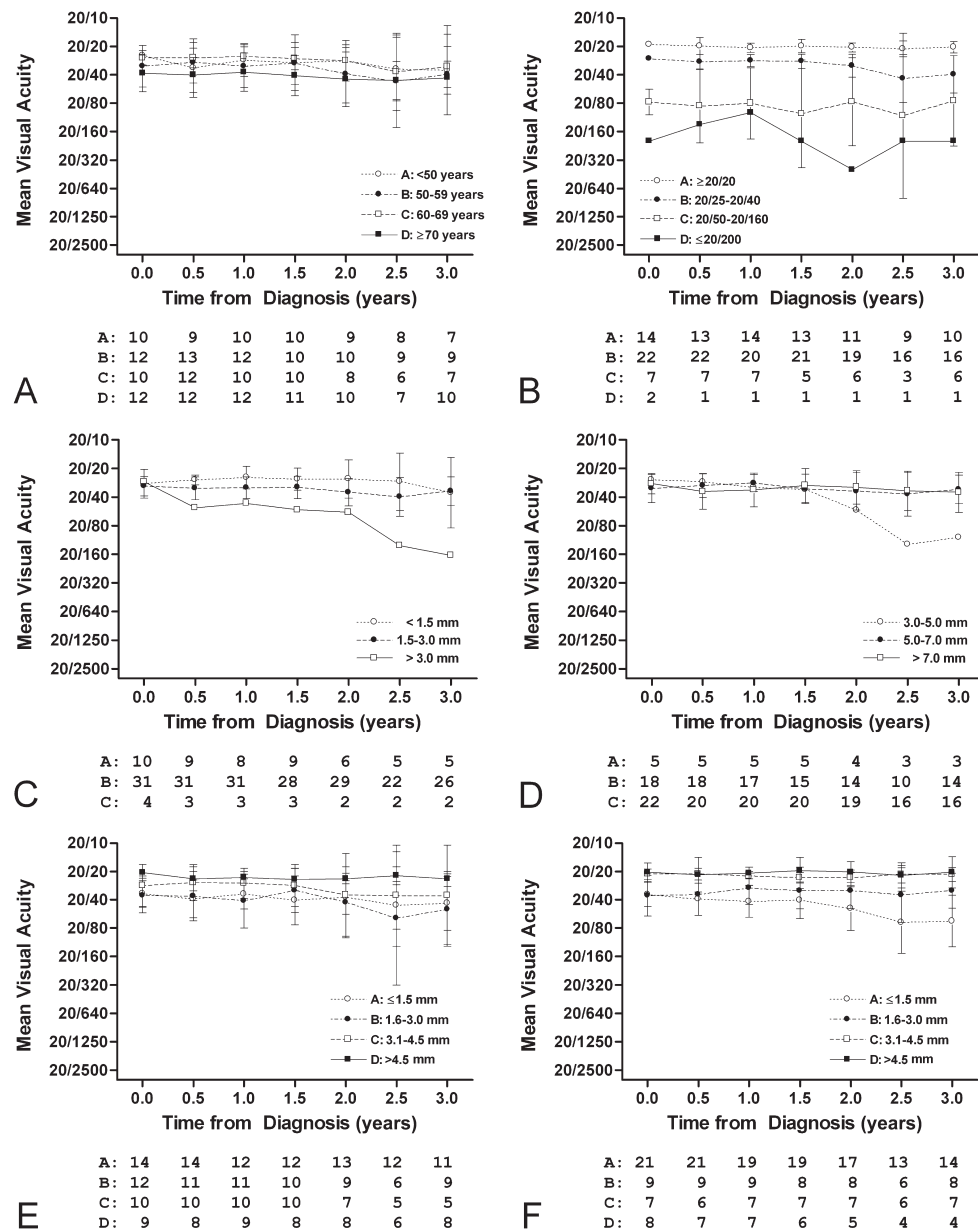
Four of 45 small choroidal melanomas <10 mm in LBD developed a local tumor recurrence after brachytherapy with a 10-mm ruthenium CCX plaque. Three of them had two or three original TFSOM high-risk characteristics for growth.² Of the 41 tumors that did not recur, 20 had 2 or 3, and 8 had 4 such characteristics. The number of high-risk characteristics for growth at diagnosis does not seem to predict local tumor recurrence after Ru-CCX brachytherapy unlike has been reported when using TTT.¹⁵ Neither were the recurring primary tumors larger than average in our series.

The planned radiation dose and treatment time were reached in each of the brachytherapies, which were followed by a local tumor recurrence. The apical radiation doses in these local recurrences were among the highest in our series. Nevertheless, because we intentionally had used in three of the four eyes with recurrence an eccentric plaque, which paralleled or did not totally cover every tumor margin, an approach that is also preferred by others for ruthenium brachytherapy when the tumor is close to the fovea,^{6,7} the scattered dose at the eccentric margin may not have been enough to effectively control the tumor contributing to local recurrence in two of these three eyes.

Tumor location likely contributed to these recurrences as well. Placing the plaque over a posterior tumor is technically challenging, and the plaque can never be placed closer than approximately 1.5 mm from the margin of the optic disk because of the optic nerve sheath, which has a diameter of 5.1 mm behind the eye.¹⁶ When managing macular tumors, the inferior oblique muscle must be detached so as not to remain between the plaque and the eye. Of the 4 recurrent tumors, 3 were located closer than 1.5 mm from the disk margin. In these cases, the plaque may have tilted,¹⁷ diminishing dose to the tumor as suggested by retrospective modeling in one of our patients, either constantly or when the patients rotate their eyes. The fourth recurrent tumor, which was not perifoveal, recurred after exceptionally not detaching the inferior oblique muscle.

We have used the 10-mm ruthenium plaque in an effort to minimize radiation-related complications when the tumor is centrally located. Although ruthenium plaques can typically be placed eccentrically,^{6,7}

Fig. 4. Mean visual acuity for 45 patients with a small choroidal melanoma irradiated with the ruthenium CCX plaque, stratified by age (A), baseline visual acuity (B), tumor thickness (C), largest basal tumor diameter (LBD) (D), and tumor distance from the optic disk (E), and foveola (F). A patient with light perception vision at the time of diagnosis was omitted from the graphs as an outlier (age, 88 years; tumor thickness, 5.2 mm; LBD, 8.0 mm; and proximity to the optic disk and the foveola 3.0 and 7.5 mm, respectively). Crossbars show 95% CIs.



round eccentric ruthenium plaques must be used with caution, especially when irradiating uveal melanomas, which are located within 1 DD from the optic disk. Only one of our 32 tumors located further than 1 DD from the disk recurred. Nevertheless, 10 of 13 tumors in our series that were located within 1 DD have not recurred locally. Likewise, tumors in contact with the optic disk have been at a high risk of local recurrence after TTT.^{15,18,19}

To the best of our knowledge, only one previous study has reported on brachytherapy for choroidal melanomas of identical size range. Semenova and Finger²⁰ treated 72 tumors that were 10 mm or smaller by LBD with palladium plaques, but they

additionally required that tumor thickness be from 1.5 mm to 2.4 mm. We irradiated 26 tumors of this size and had 1 recurrence (3.8%; 95% CI, 0–20) as compared with none (0%; 95% CI, 0–5) after palladium brachytherapy. Second, only 39 (54%) of the 72 tumors were located within 6.0 mm from the fovea and optic disk²⁰ as compared with all 26 tumors of the same thickness range in our series, a region where accurate placement of the plaque is more demanding. Finally, palladium as an isotope is biologically more effective than ruthenium, emitting gamma as compared with beta rays, which might reduce the chances of recurrence but increases the likelihood of radiation-related complications.

Transpupillary thermotherapy as a primary treatment for small choroidal melanomas is associated with a relatively high risk of local recurrence.²¹ The rate in five of seven case series we reviewed (see **Table, Supplemental Digital Content 8**, <http://links.lww.com/IAE/A249>, including summaries of material and outcomes), which best matched our series regarding tumor dimensions, was higher than in our study.^{15,19,22–24} In the largest series,¹⁸ which reported a 9% local recurrence rate at the latest follow-up similar to ours, only 30% of the tumors were perifoveal, and the mean follow-up was 19 months as compared with 60% and 62 months in our series, respectively. In fact, the Kaplan–Meier 3-year estimate of local recurrence was 22%¹⁸ as compared with 7% in our series. In 3 other TTT series, this estimate was 23% and 53% at 5 years^{15,22,24} as compared with 10% in our series. Insufficient heating of the tumor base can lead to an extrascleral recurrence after TTT,^{23,25} an alarming type of relapse that is very exceptional after brachytherapy, occurring, for example, in only 2 of 650 eyes in the COMS medium tumor trial.⁴

Any treatment of posterior choroidal melanomas affects the vision to some extent. The factors most strongly associated with a decrease in the visual acuity in our series were tumor thickness >3 mm and location within 1.5 mm from the foveola. Host-related factors such as age and visual acuity at the time of diagnosis did not predict visual loss.

The perifoveal location of most melanomas in our series explains the frequent development of maculopathy, which occurred in 49% of eyes as compared with 7% to 25% of eyes after TTT (see **Table, Supplemental Digital Content 8**, <http://links.lww.com/IAE/A248>). In these studies, the proportion of perifoveal tumors was smaller or unspecified. Transpupillary thermotherapy has little effect on vision when the tumor is not perifoveal. Immediate loss of reading vision typically occurs when the tumor is subfoveal. Maculopathy after Ru-CCX brachytherapy was typically delayed even when the tumor was subfoveal. Radiation maculopathy developed in 43% of eyes after palladium brachytherapy for choroidal melanomas of identical LBD²⁰ as compared with 31% in our entire series and only 8% in eyes with a tumor of identical thickness, although a much larger proportion of them in our series was within 6.0 mm of the optic disk and foveola.

Optic neuropathy developed unexpectedly in 1 eye in which the tumor was located 4 mm from the optic disk. Another 23 of our tumors were located within 2 DD of the optic disk, which often is considered to be the minimum safety distance for not developing radiation optic neuropathy from ruthenium brachytherapy. Juxtapapillary location thus does not seem to

be a particular risk factor for optic neuropathy when managing small uveal melanomas with the Ru-CCX plaque. In contrast, 15 of 72 (21%; 95% CI, 12–32) eyes developed radiation optic neuropathy after palladium brachytherapy,²⁰ again notwithstanding that a larger proportion of our tumors were located within 6.0 mm of the fovea and optic disk. Optic neuropathy occurred in 19% of eyes when juxtapapillary melanomas were treated with TTT,²⁶ but otherwise, the frequency has been less than 1%.¹⁸

None of the eyes developed persistent subretinal fluid or retinal detachment in our series, whereas after TTT, this has occurred in 1% to 15% of eyes.^{15,22,26} In our series, 4% of eyes developed vitreous hemorrhage, generally seen in 1% to 5% of eyes treated with TTT.^{15,19,22} Because of the posterior location of tumors in our series, anterior segment complications such as cataract, iris neovascularization, and glaucoma did not occur. Neovascular glaucoma is rare also after TTT.¹⁹

The vision improved in 51% of eyes in our series, in some patients because of cataract surgery, as is reported to occur in 6% to 30% of eyes after TTT (see **Table, Supplemental Digital Content 8**, <http://links.lww.com/IAE/A248>). Vision can also improve because of resolution of the subretinal fluid from the macula after either treatment modality.^{19,22,23,26,27} The median visual acuity at 5 years was the same as at the time of diagnosis in our series, and low vision and blindness developed in only 17% and 3% of irradiated eyes by 5 years, respectively. Thus, the long-term prognosis of preserving vision in the irradiated eye was quite favorable for most patients with a small choroidal melanoma <10 mm in LBD treated with a ruthenium-CCA plaque.

The 5-year Kaplan–Meier melanoma-related mortality estimate for small (T1a, Stage 1) tumors is 4%.¹ One patient was clinically diagnosed with possible metastatic uveal melanoma in our series, amounting to 2% mortality. Metastases after TTT of small uveal melanomas have developed in up to 5% of patients (see **Table, Supplemental Digital Content 8**, <http://links.lww.com/IAE/A248>). In 4 informative series, they developed after a local tumor recurrence in 0, 50, 75, and 100% of patients,^{15,18,19,26} as was the case with our single patient. It is not certain, however, whether metastases after a recurrence stem from the recurring tumor or whether the primary tumors are originally more aggressive, giving rise to metastasis and recurrence.

When comparing our results with those reported after TTT, it is important to take into account the many differences both in inclusion criteria and in definitions of complications. The baseline tumor characteristics and outcomes were typically more cursorily reported in the TTT studies, and survival analysis was exceptional.

A limitation of our retrospective study is that some patients had occasional missing data at individual visits to central hospitals during their later follow-up. Considering these limitations, less recurrences developed among the tumors that were managed with an Ru-CCX plaque than among patients managed with TTT and reported in the literature. In addition, the low rate of complications other than maculopathy and generally good preservation of vision might favor brachytherapy with the Ru-CCX plaque over other modalities. It remains to be studied whether this smaller plaque has an advantage over the 15-mm Ru-CCA plaque in managing small posterior choroidal melanomas <10 mm in size.

Key words: brachytherapy, choroidal melanoma, posterior melanoma, small choroidal melanoma.

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